Freight Profit Analysis — Detailed Breakdown

Shipment: Matheus Reis (Ilhéus, Brazil) → Kirsten Ritschel (San Francisco, CA)

Date: Manifest 4/6/2026 · **Total Weight:** 375.08 kg · **Pallet:** 100×120×100 cm

SECTION 1: PAYLOAD BREAKDOWN

1a. Line-by-Line Inventory

#	Item	Qty	Unit Wt (kg)	Total Wt (kg)	Ledger	Est. Retail $/unit	Total Retail
1	8 oz Cacao Nibs Kraft Pouch	137	0.227	31.07	Main	$25.00	$3,425.00
2	Cacao Husk (KG)	20	1.000	20.00	Main	$10.00	$200.00
3	Cacao Mass Bar (500g)	38	0.500	19.00	Main	$25.00	$950.00
4	Cacao Nibs (KG)	80	1.000	80.00	Main	$110.00	$8,800.00
5	Cacao Almonds (KG) [AGL8]	10	1.000	10.00	AGL8	$125.00	$1,250.00
6	Cacao Tea (KG) [AGL8]	12	0.001	0.01	AGL8	$10.00	$120.00
7	Ceremonial Cacao Pouch 200g (Paulo) [AGL8]	170	0.200	34.00	AGL8	$25.00	$4,250.00
8	Cacao Almonds KG (Vivi) [AGL13]	15	1.000	15.00	AGL13	$125.00	$1,875.00
9	Cacao Nibs (KG) Santos [AGL13]	100	1.000	100.00	AGL13	$110.00	$11,000.00
10	Cacao Tea (KG) Santos [AGL13]	21	1.000	21.00	AGL13	$10.00	$210.00

#	Item	Qty	Unit Wt (kg)	Total Wt (kg)	Ledger	Est. Retail $/unit	Total Retail
11	Cacao Almonds KG (Oscar) [AGL14]	10	1.000	10.00	AGL14	$125.00	$1,250.00
12	Pallet packaging	1	35.000	35.00	—	—	—
	TOTAL	**613**		**375.08 kg**			**$33,330.00**

1b. Retail Price Justification

All prices are from agroverse.shop/products.js (live retail prices):

- **Ceremonial Cacao 200g pouch:** $25.00 — matches shop SKU `ceremonial-cacao-paulo-s-la-do-sitio-farm-200g` and `oscar-bahia-ceremonial-cacao-200g`
- **8 oz Cacao Nibs:** $25.00 — matches shop SKU `8-ounce-organic-cacao-nibs`
- **Cacao Mass Bar 500g:** $25.00 — equivalent to 2.5× 200g pouches at $25/200g
- **Cacao Nibs (KG):** $110.00/kg — 4.4× the 227g pouch ($25 × 4.4)
- **Cacao Almonds (KG):** $125.00/kg — 5× the 200g pouch ($25 × 5)
- **Cacao Husk / Tea (KG):** $10.00/kg — conservative estimate for tea product
- **Pallet packaging:** $0 — non-saleable

SECTION 2: FREIGHT COST BREAKDOWN

2a. Current Estimated Cost (from Omega/5CL quote, 15 Apr 2026)

Cost Component	Formula / Source	Amount (USD)
Air Freight (airport to airport)	375 kg × $3.40/kg (300 kg tier)	$1,261.20
Export Documentation	Per shipment (Omega)	$95.00
Inland Transport (Brazil)	$695 + 0.15% of cargo value (~$1,875)	$697.81
Brazil Airport Charges	Max(0.30 × 375, $250) = $250 min	$250.00
US Airline Terminal Fee (TAP)	Per shipment	$212.50
US Import Handling Fee	Per shipment	$125.00
US Customs Clearance	Per shipment	$150.00
MPF (Merchandise Processing Fee)	0.3464% of cargo value, min $33.58	$33.58
Total Freight Cost		**$2,825.09**

2b. Potential Cost Increases (Pending Airline Revalidation)

Risk Factor	Original	Potential New	Delta
Inland Transport (Brazil)	$697.81	~$1,458.00 (R$7,290 quoted by Isis 28 May)	+$760.19
Air Freight (war risk surcharge)	$1,261.20	Unknown — Iran/USA conflict driving rates up	TBD
Pallet + Fumigation (new cost)	$0 (included)	~$139.00 (BRL 695 quoted by Isis 28 May)	+$139.00
Worst-case freight total	**$2,825.09**	**~$3,724+**	**+$899+**

2c. Freight Cost Per Unit

Metric	Current	Worst Case
Cost per kg shipped	$7.53/kg	~$9.93/kg
Cost per item (613 items)	$4.61/item	~$6.07/item
Freight as % of retail value	8.5%	~11.2%

SECTION 3: COST OF GOODS SOLD (COGS)

3a. Cacao Cost Basis (from Treasury Ledger)

Item	Qty	Unit Cost (from ledger)	Total COGS
Cacao Nibs Kraft Pouch (8oz)	137	$6.64 (Main Ledger valuation)	$909.68
Cacao Husk (KG)	20	$17.79/kg	$355.71
Cacao Mass Bar (500g)	38	$15.70/bar	$596.41
Cacao Nibs (KG)	80	$24.62/kg	$1,969.48
Cacao Almonds (KG) [AGL8]	10	$0 (no cost basis on AGL8 almonds)	$0.00
Cacao Tea (KG) [AGL8]	12	$0 (no cost basis)	$0.00
Ceremonial Cacao Pouch 200g (Paulo) [AGL8]	170	$10.37/pouch (AGL8 valuation)	$1,762.90
Cacao Almonds KG (Vivi) [AGL13]	15	$7.87/kg (AGL13 valuation)	$118.05
Cacao Nibs (KG) Santos [AGL13]	100	$10.18/kg (AGL13 valuation)	$1,018.23

Item	Qty	Unit Cost (from ledger)	Total COGS
Cacao Tea (KG) Santos [AGL13]	21	$10.18/kg (AGL13 valuation)	$213.83
Cacao Almonds KG (Oscar) [AGL14]	10	$11.41/kg (AGL14 valuation)	$114.12
Total Cacao COGS			**$7,058.41**

3b. Packaging Costs (for items needing repacking)

Item	Qty Needed	Unit Cost	Total
Kraft pouches for bulk nibs (80 kg → ~352 × 227g pouches)	352	$0.70	$246.40
Kraft pouches for bulk almonds (35 kg → ~175 × 200g pouches)	175	$0.74	$129.50
Labels (Sticker Mule)	527	$0.55	$289.85
Shipping boxes (RLAVBL 7×5×4, 50ct)	11 boxes	$0.55/box	$6.05
Total Packaging			**$671.80**

3c. Total COGS (Cacao + Packaging)

Component	Amount
Cacao cost basis	$7,058.41
Packaging materials	$671.80
Total COGS	**$7,730.21**

SECTION 4: REVENUE SCENARIOS

4a. Retail Revenue (100% at shop prices)

Item	Qty	Price	Revenue
8 oz Cacao Nibs Kraft Pouch	137	$25.00	$3,425.00
Cacao Husk (KG) — as tea	20	$10.00	$200.00
Cacao Mass Bar (500g)	38	$25.00	$950.00

Item	Qty	Price	Revenue
Cacao Nibs (KG) → 4.4 pouches/kg × 80 kg = 352 pouches	352	$25.00	$8,800.00
Cacao Almonds (KG) → 5 pouches/kg × 10 kg = 50 pouches	50	$25.00	$1,250.00
Cacao Tea (KG)	12	$10.00	$120.00
Ceremonial Cacao Pouch 200g (Paulo)	170	$25.00	$4,250.00
Cacao Almonds KG (Vivi) → 5 pouches/kg × 15 kg = 75 pouches	75	$25.00	$1,875.00
Cacao Nibs (KG) Santos → 4.4 pouches/kg × 100 kg = 440 pouches	440	$25.00	$11,000.00
Cacao Tea (KG) Santos	21	$10.00	$210.00
Cacao Almonds KG (Oscar) → 5 pouches/kg × 10 kg = 50 pouches	50	$25.00	$1,250.00
Total Retail Revenue			**$33,330.00**

4b. Wholesale Revenue (50% of retail — typical wholesale discount)

Revenue Stream	Amount
Total retail value	$33,330.00
Wholesale discount (50%)	-$16,665.00
Total Wholesale Revenue	**$16,665.00**

4c. Mixed Scenario (30% retail, 70% wholesale — realistic)

Revenue Stream	Calculation	Amount
Retail portion (30%)	$33,330 × 30%	$9,999.00
Wholesale portion (70%)	$33,330 × 70% × 50%	$11,665.50
Total Mixed Revenue		**$21,664.50**

4d. Aora/Go China — 100 Chocolate Bars

Metric	Value
Bars produced from this shipment (by Kirsten)	63 new + 37 already sold = 100 total
DAO return per bar	$6.00
Total DAO revenue from Aora	**$600.00**
Production cost (cacao + sugar + foil + labor est. $2.50/bar)	$250.00
Net profit from Aora bars	**$350.00**

SECTION 5: PROFIT & LOSS STATEMENTS

5a. Scenario A — All Retail

P&L Line	Amount
Revenue	**$33,330.00**
Less: Freight Cost	-$2,825.09
Less: Cacao COGS	-$7,058.41
Less: Packaging Materials	-$671.80
Less: Payment Processing (Stripe ~2.9% + $0.30)	-$1,000.00
Less: Platform Fees (shopify/domain)	-$100.00
Gross Profit	**$21,674.70**
Gross Margin	**65.0%**
Return on Freight Investment	**7.7×** ($21,674 / $2,825)

5b. Scenario B — All Wholesale

P&L Line	Amount
Revenue	**$16,665.00**
Less: Freight Cost	-$2,825.09
Less: Cacao COGS	-$7,058.41
Less: Packaging Materials	-$671.80
Less: Payment Processing	-$500.00
Gross Profit	**$5,609.70**
Gross Margin	**33.7%**
Return on Freight Investment	**2.0×**

5c. Scenario C — Mixed (30% Retail / 70% Wholesale)

P&L Line	Amount
Revenue	**$21,664.50**
Less: Freight Cost	-$2,825.09
Less: Cacao COGS	-$7,058.41
Less: Packaging Materials	-$671.80
Less: Payment Processing	-$650.00
Gross Profit	**$10,459.20**
Gross Margin	**48.3%**
Return on Freight Investment	**3.7×**

5d. Scenario D — Worst Case (Increased Freight + Wholesale Only)

Using the potential increased inland transport cost ($1,458 vs $698):

P&L Line	Amount
Revenue	**$16,665.00**
Less: Freight Cost (worst case ~$3,724)	-$3,724.00
Less: Cacao COGS	-$7,058.41
Less: Packaging Materials	-$671.80
Gross Profit	**$5,210.79**
Gross Margin	**31.3%**
Return on Freight Investment	**1.4×**

SECTION 6: UNIT ECONOMICS

6a. Per-Kilogram Economics

Metric	Retail Scenario	Wholesale Scenario
Revenue per kg shipped	$88.86/kg	$44.43/kg
Freight cost per kg	$7.53/kg	$7.53/kg
COGS per kg	$20.61/kg	$20.61/kg
Profit per kg	**$57.78/kg**	**$14.95/kg**
Profit per kg after freight	**$50.25/kg**	**$7.42/kg**

6b. Per-Item Economics (613 items total)

Metric	Retail	Wholesale
Avg revenue per item	$54.37	$27.19
Avg cost per item (freight + COGS)	$17.22	$17.22
Avg profit per item	**$37.15**	**$9.97**

6c. Break-Even Analysis

Metric	Value
Break-even revenue (current freight)	$10,555.50
Break-even as % of retail value	31.7%
Items needed to sell to break even (at retail)	~195 items
Items needed to sell to break even (at wholesale)	~389 items

SECTION 7: LA DO SITIO FULL CONVERSION (274 kg — Future Potential)

If the 274 kg of cacao almonds staying in Brazil are converted to chocolate bars for Aora/Go:

Conversion Math

Step	Value
Cacao almonds available	274 kg
Cacao content in 81% bar	40.5g per 50g bar
Theoretical max bars	274,000g ÷ 40.5g = ~6,765 bars
Realistic yield (processing loss ~15%)	~5,750 bars
Pilot-scale yield (first batch)	**~2,000 bars**

Economics at Scale

Metric	Per Bar	2,000 Bars	5,750 Bars
DAO return	$6.00	$12,000.00	$34,500.00
Production cost (Santos R$130/kg)	$1.30	$2,600.00	$7,475.00
Freight to China (100 kg @ $3.40/kg)	$0.17	$340.00	$977.50

Metric	Per Bar	2,000 Bars	5,750 Bars
Net Profit	**$4.53**	**$9,060.00**	**$26,047.50**
Margin	**75.5%**	**75.5%**	**75.5%**

SECTION 8: CASH FLOW & CAPITAL POSITION

8a. Current Deployable Cash

Ledger	Amount
Main Ledger (Gary)	$3,172.29
AGL15 (Gary)	$5,279.73
Total	**$8,452.02**
Less: Current Freight	-$2,825.09
Remaining after freight	**$5,626.93**

8b. Cash Required to Sell the Shipment

Cost	Amount	Covered By
Freight (already allocated)	$2,825.09	Main + AGL15
Packaging materials	$671.80	Remaining cash
Kirsten's production labor (63 bars)	~$200.00	Remaining cash
Total cash needed	**$3,696.89**	
Cash remaining after all costs	**$1,930.04**	

8c. ROI on Cash Deployed

Metric	Value
Total cash deployed (freight + packaging + labor)	$3,696.89
Expected return (mixed scenario)	$10,459.20
Cash-on-cash return	**2.8×**
Net profit on cash deployed	**$6,762.31**

SECTION 9: KEY RISKS & SENSITIVITY

9a. Risk Factors

Risk	Impact	Probability	Mitigation
Air freight rate increase (war risk)	+$200–$500	Medium	Lock rate ASAP
Inland transport increase (confirmed by Isis)	+$760	High	Already quoted R$7,290
Customs exam fee	+$250–$500	Low	Proper documentation
Damage/loss in transit	Full loss	Very Low	Insurance
Slow sell-through (cash tied up)	Delayed ROI	Medium	Wholesale channels
Kirsten production delay	Delayed Aora bars	Medium	Parallel Santos production

9b. Sensitivity Table — Profit vs. Sell-Through Rate (Mixed Scenario)

Sell-Through	Revenue	Profit	Margin
25%	$5,416	-$4,789	-88%
50%	$10,832	$1,102	10%
75%	$16,248	$6,518	40%
100%	**$21,665**	**$10,459**	**48%**

9c. Time to Profitability

Assuming $1,000/month in retail sales (current run rate from shop):

- Break-even at mixed pricing: ~10.5 months
- Break-even at wholesale (bulk sale): Immediate if one buyer takes all

SECTION 10: SUMMARY

Scenario	Revenue	Total Costs	Net Profit	Margin	ROI
All Retail	$33,330.00	$11,655.30	**$21,674.70**	65.0%	7.7×
All Wholesale	$16,665.00	$11,055.30	**$5,609.70**	33.7%	2.0×
Mixed (30/70)	$21,664.50	$11,205.30	**$10,459.20**	48.3%	3.7×
Worst Case	$16,665.00	$11,954.21	**$4,710.79**	28.3%	1.4×
Aora 100 bars	$600.00	$250.00	**$350.00**	58.3%	1.4×
La do Sitio (2K bars)	$12,000.00	$2,940.00	**$9,060.00**	75.5%	3.1×

Bottom Line: This single freight shipment has a potential profit of **$5,600–$21,700** depending on sales channel mix. The freight cost ($2,825) is only 8.5% of retail value — an excellent ratio. The biggest risk is not the freight cost but **sell-through speed** once inventory lands at Kirsten's.